Correspondence

A Robust Adaptive Quantizer

DAVID J. GOODMAN AND ROGER M. WILKINSON

Abstract—**A popular method of time-varying quantization involves a multiplicative change in quantizer overload point after each sample time. Although this technique produces an average loading factor that is invariant with rms input, it is vulnerable to transmission errors. Here, we introduce a related quantizer which has the advantage of disipating the effects of transmission errors at any desired rate. On the other hand, adaptation is not perfect. The loading factor now depends on rms input and in designing the quantizer one must compromise between the goal of constant loading factor and the goal of rapid error dissipation. We provide a design method and apply it to the modification of a quantizer currently used for speech. The new quantizer dissipates transmission errors with a 5 ms time constant while maintaining accurate adaptation over a 40 dB input range.**

I. ADAPTIVE QUANTIZATION

The past two years have seen the appearance of several papers describing differential pulse code modulation (DPCM) encoders with time-varying quantizers [1]–[5]. In all of them, the quantizer characteristic is controlled by the output code sequence with the aim of maintaining a constant loading factor. The mechanism may be viewed as a form of automatic gain control with feedback around the quantizer, as shown in Fig. 1. Here we have the analog input, z, multiplied by $1/x$ and processed by a quantizer with a 1-V overload point. If at the digital-to-analog converter, the normalized output is multiplied by x, the overall effect is that of processing z with the quantizer of Fig. 2, in which the overload voltage is x. If σ is the rms input, the loading factor is $y = x/\sigma$. Given a quantizer shape and information about z, a designer can specify a desired loading factor, $\hat{y}$, which provides an appropriate mixture of granular quantizing noise and overload distortion.

In speech telephony, the locally measured rms input level fluctuates with time about a long-term average that depends on the talker and his communication path to the encoder. With a fixed quantizer, x is constant and y is inversely proportional to σ which can vary over as much as 40 dB (exclusive of silent intervals). In an adaptive quantizer, x varies with time in a manner that reduces the dependence of y on σ.

The signal that controls x contains statistical information about σ so that even with σ constant, x is necessarily a random variable, fluctuating around a central value, $\bar{x}$. It follows that a reasonable design goal for the adaptive quantizer is to make $\bar{y}$, the steady-state loading factor (defined as $\bar{x}/\sigma$), equal to $\hat{y}$, the desired loading factor. It has been shown [4], [6] that the quantizers described in the recent papers can adapt to any rms input in the sense that $\bar{y}$ remains independent of σ over any desired range of input levels. In building the quantizer, one must establish maximum and minimum values of x, causing $\bar{y}$ to behave as in Fig. 3, which also shows the relationship of y to σ in a time-invariant quantizer.

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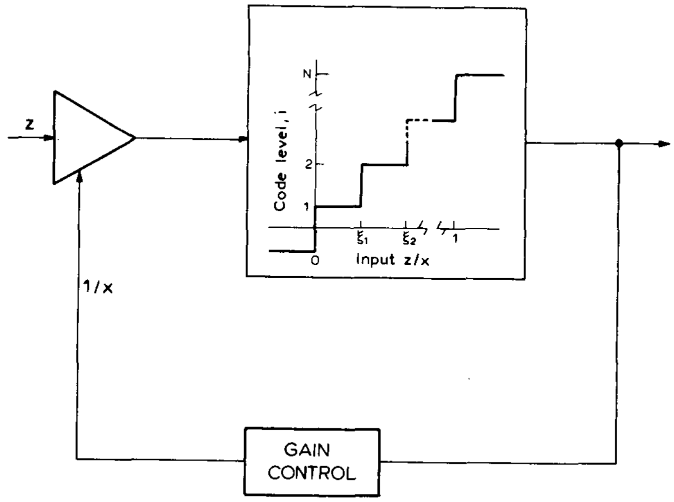

Fig. 1. Adaptive quantization.

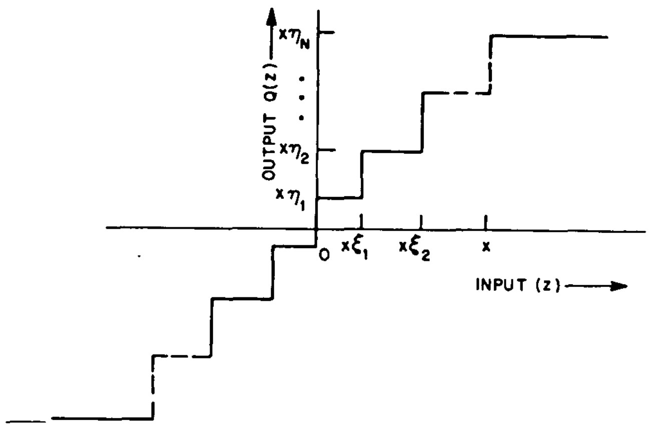

Fig. 2. Quantizer characteristic.

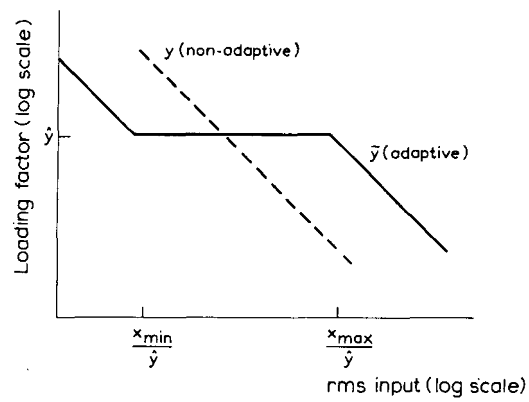

Fig. 3. Dynamic range without error dissipation.

A. The Adaptation Rule

In [1]–[5], the quantizer overload point, x changes at each sample time by a multiplicative quantity that depends only on the most recently generated code word. The condition for finding z_k in the ith-positive or the ith-negative quantizing interval is

$$x_k\xi_{i-1} \leq |z_k| < x_k\xi_i, \tag{1}$$

where x_k is the overload point at time k.

In this case the quantizer output is $Q(z_k) = \pm x_k\eta_i$ and the magnitude of the transmitted code word is $I_k = i$. Before z_{k+1} is coded, the overload point changes from x_k to $M(i)x_k$, or, in general[1]

$$x_{k+1} = M(I_k)x_k. \tag{2}$$

In a quantizer with $2N$ output levels $i = 1,2,\cdots,N$, the occurrence of an "inner" code word ($I_k = i$ near 1) suggests that x_k is too large and should be decreased. Conversely an "outer" code word (i near N) suggests that the quantizer may be overloaded and that x_k should be increased. It follows that the N multipliers should satisfy

$$M(1) < 1, \qquad M(N) > 1$$

$$M(1) \leq M(2) \leq \cdots \leq M(N).$$

B. Vulnerability to Transmission Errors

The title of [1] contains the phrase "one word memory" and it is important to appreciate that the memory referred to is the number of storage registers required by the algorithm. In the information-theory sense, the memory is infinite because each overload voltage, x_k, depends on the entire past of the code sequence

$$x_{k+1} = \prod_{m=0}^{k} M(I_m)x_0. \tag{4}$$

This equation exposes a disadvantage of the algorithm, its sensitivity to transmission errors. Let x' and I' be the decoder versions of x and I, respectively and assume that at time $l < k$, $I_l = i$ while a transmission error causes $I_l' = j$. If there are no other errors,

$$x_{k+1}' = \frac{M(j)}{M(i)}\, x_{k+1}. \tag{5}$$

Each error causes a multiplicative offset between receiver and transmitter that in principle can persist indefinitely.

II. A ROBUST QUANTIZER

Our purpose in this correspondence is to investigate the following adaptation procedure in which the effect of a transmission error diminishes with time

$$\begin{aligned} x_{k+1} &= M(I_k)x_k^{\beta} \\ &= \prod_{m=0}^{k} [M(I_m)]^{\beta^{(k-m)}} x_0^{\beta^{k+1}}. \end{aligned} \tag{6}$$

If at time l, j is received instead of i,

$$\frac{x_{k+1}'}{x_{k+1}} = \left[\frac{M(j)}{M(i)}\right]^{\beta^{(k-l)}}. \tag{7}$$

When $\beta < 1$, the offset due to each error decays exponentially with time, causing the quantizer to be more robust in the presence of transmission errors than a quantizer operating according to (2), where implicitly $\beta = 1$.

The effect of $\beta < 1$ is analogous to that of a leaky integrator in DPCM. To establish the analogy, let $u = \log x$ and $m = \log M$. In terms of the logarithmic quantities

$$u_{k+1} = \beta u_k + m(i) \text{ when } \xi_{i-1} \leq |z_k|/x_k < \xi_i. \tag{8}$$

Fig. 4 shows an adaptive quantizer that functions according to (8). The integrator time constant RC is related to β by

[1] In [1], [2], [5] this algorithm is stated explicitly, while in [3], [4] it is implied by the exponentially weighted estimation procedures. In the Appendix, we derive $M(i)$ as a function of the estimation parameters in [3] and [4].

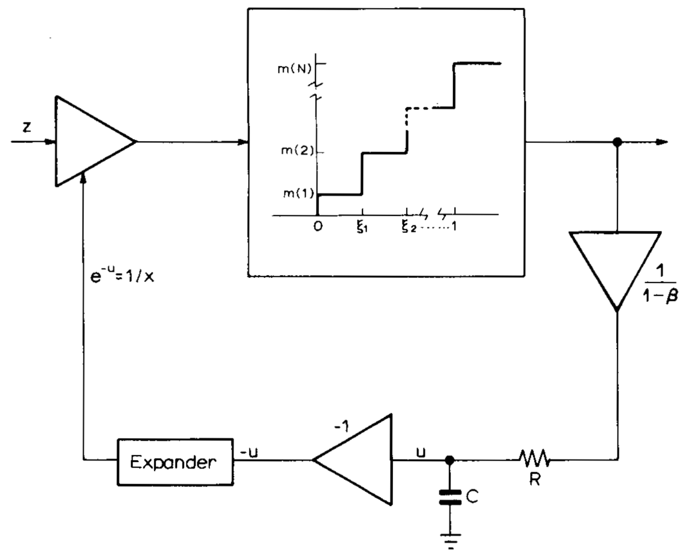

Fig. 4. Robust quantizer.

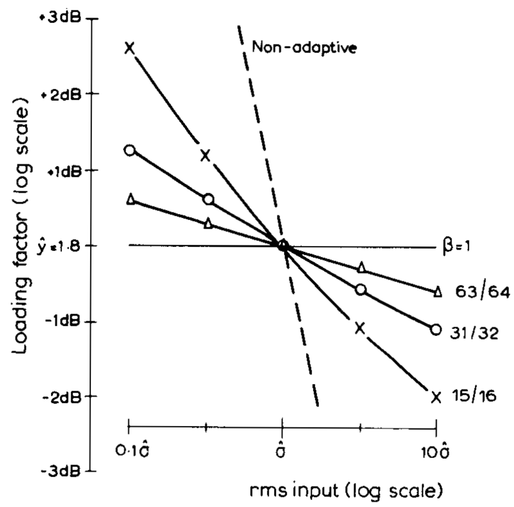

Fig. 5. Dynamic range of robust quantizer.

$$\beta = \exp(-T/RC) \tag{9}$$

where T is the sample time of the encoder.

III. PRINCIPAL RESULTS

A. Performance

Following the static-range analysis of [6], we have derived an equation that relates $\bar{y}$, the steady-state loading factor, to the multipliers $M(i)$ and the probability distribution of an independent, identically distributed input sequence $\{z_k\}$. The equation shows that the price of the error dissipation provided by (7) is the constant loading factor $\bar{y}$. With $\beta < 1$, $\bar{y}$ varies with σ as in Fig. 5, where we see that as β is adjusted downward from unity, the dependence on σ increases. On the other hand, low β means rapid error dissipation. Consequently, the design value of β is a compromise between sensitivity of the average loading factor to changes in rms input level and sensitivity of the receiver to the effects of transmission errors.

B. Design

The multipliers, $M(i)$, of the robust quantizer may be derived from a set, $M'(i)$, that has been designed for $\beta = 1$. The design of $M'(i)$ may be empirical as in [1]–[5] or it may proceed from the

TABLE I
COMPARISON OF ERROR DISSIPATION FACTORS (β)

		Error Dissipation Offset 1 dB		Loading Factor		Decibels Relative to $\hat{y}$	
β	RC (ms)	Samples	Time (ms)	$\sigma = 10\,\hat{\sigma}$	$\sigma = 0.1\,\hat{\sigma}$	$\sigma = 10\,\hat{\sigma}$	$\sigma = 0.1\,\hat{\sigma}$
15/16	2.6	38	6.3	1.42	2.43	−2.02	2.60
31/32	5.2	77	12.8	1.58	2.08	−1.10	1.26
63/64	10.6	155	25.8	1.68	1.93	−0.58	0.62

$M'(i)$ as in (10), 6 kHz sampling.

design formulae of [6] which require assumptions about signal statistics and specifications of static and dynamic performance goals. In the robust quantizer, one wishes to specify, in addition to $\hat{y}$, the rms level at which $\bar{y} = \hat{y}$. This level, $\hat{\sigma}$, shown in Fig. 5, will be within the range of anticipated quantizer input levels. When $\sigma > \hat{\sigma}, \bar{y} < \hat{y}$ and conversely. If the multipliers $M'(i)$ lead to the constant loading factor, $\hat{y}$, when $\beta = 1$, the multipliers

$$M(i) = (\hat{\sigma}\hat{y})^{1-\beta} M'(i)$$

will give $\bar{y} = \hat{y}$ when $\sigma = \hat{\sigma}$ and $\beta < 1$.

C. An Example

Fig. 5 has been obtained for the 16-level quantizer described in [7] with multipliers

$$M'(1) = M'(2) = M'(3) = M'(4) = 0.8;$$

$$M'(5) = M'(6) = 1.25; \quad M'(7) = 2; \quad M'(8) = 3. \qquad (10)$$

If the quantizer input is Gaussian, (12) of Section IV with $\beta = 1$ has the solution $\hat{y} = 1.8$, the nominal loading factor for this quantizer. In Fig. 5, we show for $\beta = 15/16, 31/32, 63/64$, the dependence of $\bar{y}$, the actual loading factor, on rms input over the 40 dB range 0.1 $\hat{\sigma}$ to 10 $\hat{\sigma}$. What is an appropriate design value for β? The most damaging transmission errors cause x' at the receiver to be multiplied by $M'(8) = 3$ when x at the transmitter is multiplied by $M(1) = 0.8$. The initial offset is $3/0.8 = 11.5$ dB. Assume that such an error occurs at $k = 0$, and find the number of samples, n, that elapse before the offset x'/x is reduced to 1 dB. From (7) we find that n satisfies $20 \log (x_n'/x_n) = \beta^n 20 \log (3/0.8) = 1$ or $\beta^n = 1/11.5$. Table I summarizes the tradeoff between error dissipation time n and loading factor variations. With 6 kHz sampling and $\beta = 31/32$, the error dissipation time constant is 5.2 ms and the offset is reduced to 1 dB within 13 ms. In addition, the loading factor is confined to ± 1.26 dB of the nominal value 1.8.

It remains to establish the voltage range over which the quantizer operates. Designing for a peak speech voltage of 5 V, it is reasonable to anticipate a peak quantizer input (difference between the speech signal and DPCM feedback voltage) of 2.5 V. Thus at the upper end of the adaptation range ($\sigma = 10\,\hat{\sigma}$) we set the quantizer overload point $x = 10\,\hat{\sigma}\bar{y} = 2.5$. For this input, Table I shows $\bar{y}$ to be 1.58 which implies $\hat{\sigma} = 0.16$ V and with $\beta = 31/32$

$$M(i) = (1.8 \times 0.16)^{1/32} M'(i) = 0.96 M'(i).$$

IV. ANALYSIS

To find the average loading factor as a function of signal statistics and multipliers, we refer to [6], in which the steady-state overload point $\bar{x}$ is defined as the value of x for which the expected change in $\log x$ is zero. When $x_k > \bar{x}, x_{k+1}$ is, on the average, less than x_k and conversely.

Assume that the input sequence $\{z_k\}$ is independent, identically distributed and that the shape of the probability distribution is known, but not the variance. That is, consider

$$\varphi(w) = \text{Prob}\,[|z_k|/\sigma < w]$$

to be given. The expected change in $\log x$ will be the sum of the changes associated with each output level, i, weighted by Prob $[I_k = i \mid x_k = x]$. This is the probability that $|z_k|$ is in the range given in (1) when $x_k = x$, which is

$$p_i(x/\sigma) = \varphi(\xi_{i-1}x) - \varphi(\xi_i x).$$

Thus we have from (6)

$$E[\log x_{k+1} - \log x_k \mid x_k = x] = \sum_{i=1}^{N} p_i(x/\sigma) \cdot [\log M(i) + \beta \log x - \log x]. \qquad (11)$$

If we let $y = x/\sigma$, and observe that $\bar{y} = \bar{x}/\sigma$ is the loading factor for which (11) is zero, we have $\bar{y}$ as the solution to

$$B(y) = \sum_{i=1}^{N} p_i(y) \log M(i) = (1 - \beta) \log (\sigma y). \qquad (12)$$

$B(y)$ is the adaptation bias function which in [6] is shown to be monotone decreasing from $\log M(N) > 0$ to $\log M(1) < 0$ as shown in Fig. 6. The root of $B(y) = 0$, $\hat{y}$, is the (constant) average loading factor when $\beta = 1$. It is also the loading factor for $\beta < 1$ and $\sigma = 1/\hat{y}$, because the right side of (12) is zero when $\sigma\hat{y} = 1$. In Fig. 6(b), we show as functions of y: $\log(\sigma_1 y)$, $\log(\sigma_2 y)$ and $B(y)/(1 - \beta_1)$, where $\sigma_1 < 1/\hat{y} < \sigma_2$ and $\beta < 1$. At the points of intersection we find $\bar{y}_1$ and $\bar{y}_2$, the average loading factors for σ_1 and σ_2. In Fig. 6(c) we have $B(y)/(1 - \beta_2)$, $\beta_1 < \beta_2 < 1$ and $\bar{y}_1$ and $\bar{y}_2$ closer together than in Fig. 6(b). Hence the shape of Fig. 5, which shows the dependence of $\bar{y}$ on σ to be small when $\beta \approx 1$.

V. CONCLUSION

The error-dissipating mechanism mitigates a serious disadvantage of a promising coding technique. On the other hand it degrades to some extent the performance of a coder operating with an error-free channel. Although the theoretical work presented here suggests that this degradation can be limited, it would be desirable to have an empirical comparison of improvements under noisy channel conditions and penalties in the error-free situation.

Fig. 4 may be viewed as a means of quantizer realization and in fact it is a generalization of a circuit that appears in a 2-bit adaptive PCM encoder [8]. Although Fig. 4 shows an analog implementation, digital hardware may also be used. In a digital circuit, a first-order recursive filter operating according to (8) would replace the RC filter and the amplifier with gain $(1 - \beta)^{-1}$. With the values of β in Table I and Fig. 5, the digital filter multiplication can be realized as a shift and one subtraction per sampling interval.

APPENDIX

In [3], [4] the adaptive quantizer is formulated as a device that estimates the rms input from the sequence of quantizer outputs. The estimation is sequential with exponential weighting. If $\{q_k\}$ is

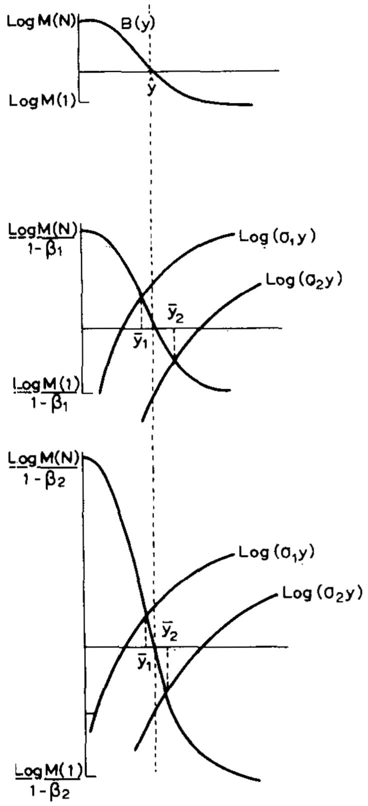

Fig. 6. Average loading factors.

the sequence of output levels ($q_k = x_k\eta_i$ when $I_k = i$) the variance estimate is

$$S_{k+1}^2 = (1-\alpha)\sum_{m=0}^{k}\alpha^m q_{k-m}^2 = (1-\alpha)q_k^2 + \alpha S_k^2. \quad \text{(A1)}$$

The adaptation makes the quantizer overload point proportional to S:

$$x_{k+1} = \hat{y}S_{k+1} = [(1-\alpha)q_k^2 + \alpha S_k^2]^{1/2}\hat{y}. \quad \text{(A2)}$$

By substituting $x_k\eta_i$ for q_k and $x_k/\hat{y}$ for S_k in (A2) we obtain

$$x_{k+1} = x_k[\hat{y}^2(1-\alpha)\eta_i^2 + \alpha]^{1/2}$$

which is equivalent to (2) with

$$M(i) = [\hat{y}^2(1-\alpha)\eta_i^2 + \alpha]^{1/2}. \quad \text{(A3)}$$

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Long Frame Sync Words for Binary PSK Telemetry

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***Abstract*—Correlation criteria have previously been established for identifying whether a given binary sequence would be a good frame sync word for phase-shift keyed telemetry. In the past, the search for a good K-bit sync word has involved the application of these criteria to the entire set of 2^K binary K-tuples. It is shown that restricting this search to a much smaller subset consisting of K-bit prefixes of pseudonoise sequences results in sync words of comparable quality, with greatly reduced computer search times for larger values of K. As an example, this procedure is used to find good sync words of length 16–63; from a storage viewpoint, each of these sequences can be generated by a 5- or 6-bit linear feedback shift register.**

I. BACKGROUND

Consider the problem of determining a suitable frame sync word for a digital communication link employing binary phase-shift keyed (PSK) modulation. For uncoded data received over the additive white Gaussian noise channel, the optimum procedure for locating the received sync words is based on a bit correlation rule [1]. In the block-coded case, the optimum sync word search requires a word correlation rule [2]; however, practical considerations often dictate the use of the suboptimal bit correlation approach.

For example, if format constraints require that the sync word not contain an integral number of block code words, the bit correlation scheme is much less complex than the optimum frame sync rule [2, p. 1130]. In the case of convolutionally coded telemetry as well, the suboptimal bit correlation sync word search is desirable from a complexity viewpoint.

Suppose the sync word is a K-bit sequence of 1's and 0's, $s = (s_0, s_1, \cdots, s_{K-1})$, with the nonperiodic [3, p. 195] autocorrelation function

$$C_l = K - l - 2\sum_{i=0}^{K-l-1}(s_i \oplus s_{i+l}), \qquad 0 \le l \le K. \quad (1)$$

In a binary PSK modulation system, because the derived carrier reference in the receiver has an inherent 180° phase ambiguity, the detected data stream will be inverted with probability $\frac{1}{2}$ (e.g., [4]). Consequently, when a bit correlation sync rule is used, the probability of false synchronization is minimized by selecting a sync word for which $|C_l|$ is minimized over the range $1 \le l \le K-1$. That is, ideally the autocorrelation function should have the property

$$|C_l| \le 1, \qquad 1 \le l \le K-1. \quad (2)$$

Sequences with sidelobe correlations satisfying this constraint are called Barker codes [5]; they are optimum sync words for the channels of interest. Unfortunately, it is believed that no binary sequences of length greater than 13 satisfy the Barker restriction of (2) (e.g., [6]).

Generally, if sync words with suitable autocorrelation functions are considered, frame sync performance improves exponentially with increasing length K. For $K > 13$, the Barker constraint must be relaxed somewhat, although it is still desirable that $|C_l|$ be near zero for $l \ne 0$. Define the maximum sidelobe correlation

$$C_{\max} \equiv \max\{|C_l|: 1 \le l \le K-1\} \quad (3)$$

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